

# **IF 260**

## **Sistem Operasi**

### **Minggu 11**

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# Agenda

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- **Review:**
- **Topik Minggu 11:**
  - **Storage Media (Media Penyimpanan Sekunder)**
  - **File Systems (Sistem-sistem Berkas)**

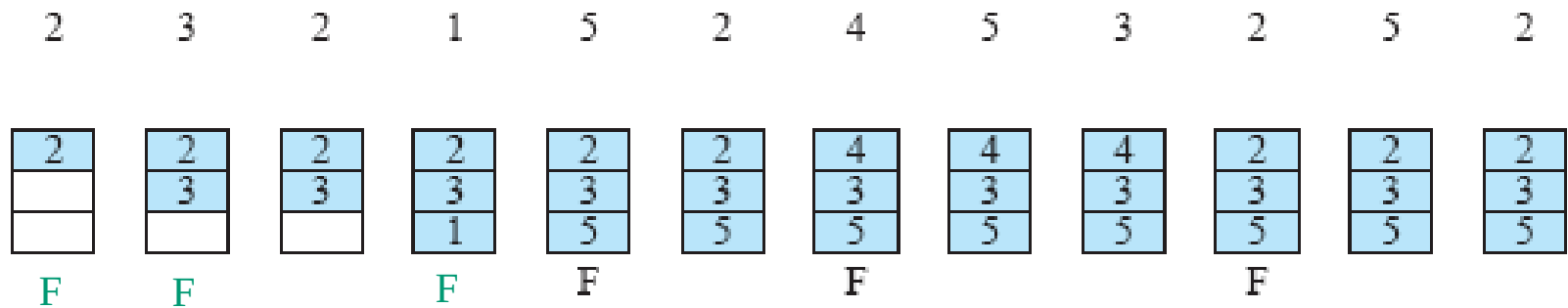
# **Review:**

## **Page Replacement Algorithms**

# Optimal Page Replacement Algorithm (Belady's Algorithm)

- Memilih page yang nantinya akan diacu paling terakhir untuk dikeluarkan
  - Optimal (**page fault paling rendah**), tetapi tidak realistik karena mensyaratkan OS punya informasi sempurna akan event-event yang akan terjadi di depan

- Contoh:



- Total page-fault = 6
- Jumlah page-fault untuk penggantian page = 3
- Untuk slide2x selanjutnya, fokus pada page-fault untuk penggantian page untuk membandingkan page replacement algorithm

# First-In, First-Out (FIFO) Page Replacement Algorithm

- Page frames yang lebih dahulu di-loaded ke memory yang akan lebih dahulu dikeluarkan apabila ada page-fault
  - Keuntungan: sederhana untuk diimplementasikan
  - Kekurangan: asumsi bahwa page yang paling lama berada di memory akan paling kecil peluangnya untuk diacu sering kali salah → kinerja algoritma ini kurang bagus

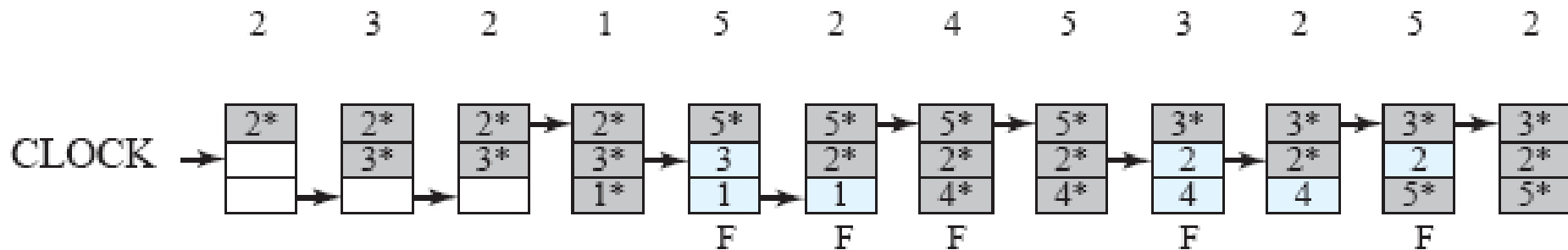
- Contoh:

|                                                                        |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 2                                                                      | 3 | 2 | 1 | 5                                                                       | 2 | 4 | 5 | 3                                                                       | 2 | 5 | 2 |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 1                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 1                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 4                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 4                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 4                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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|                                                                        |   |   |   | F                                                                       | F | F |   | F                                                                       |   | F | F |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |

- Jumlah page-fault untuk penggantian page = 6

# Clock Page Replacement Algorithm: Contoh

- Contoh:



- Bagaimana gambaran clock kalau mengacu ke slide sebelumnya?
- Jumlah page fault untuk penggantian page = 5

# Least-Recently Used (LRU) Page Replacement Algorithm

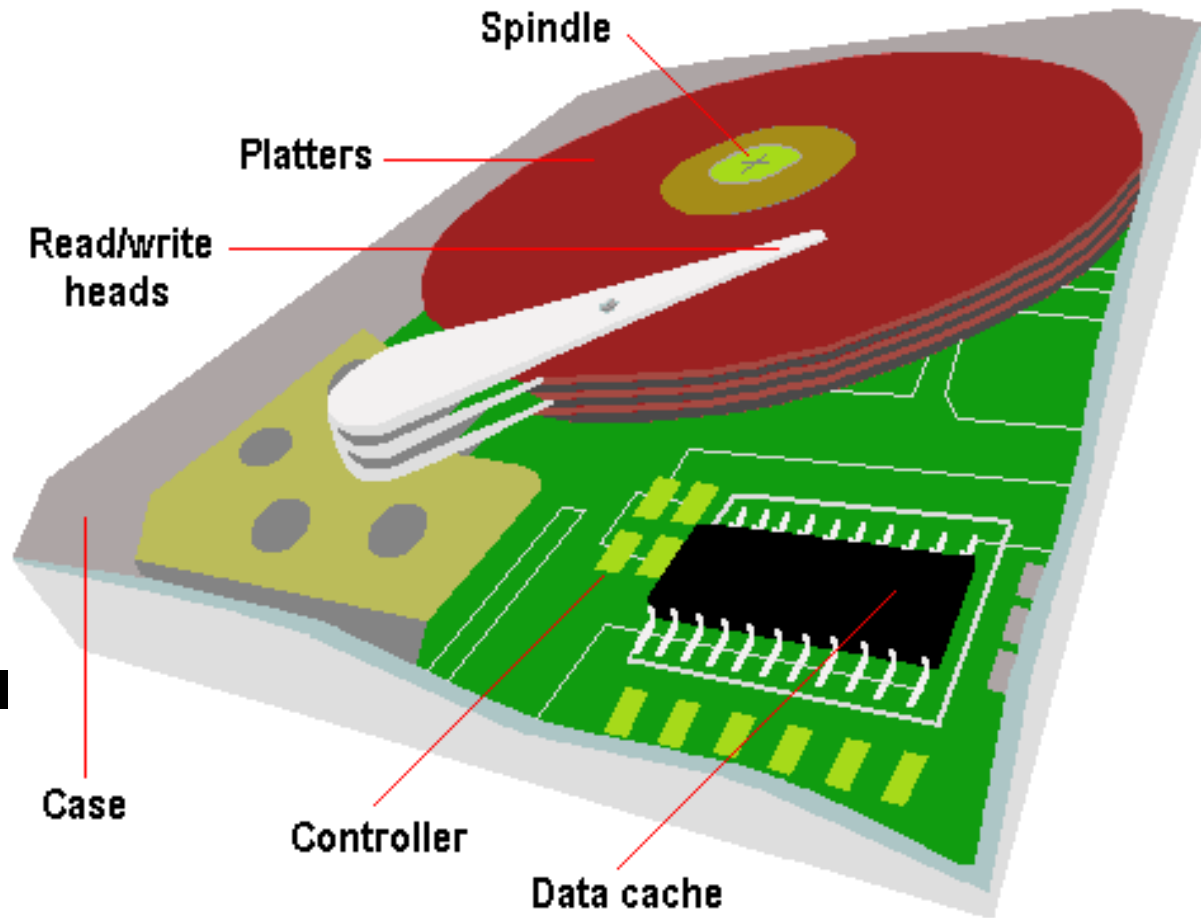
- Keluarkan page yang paling lama tidak terpakai
- Mendekati optimal algorithm, namun implementasi lebih rumit dan lebih memakan waktu
  - Mengatur pengurutan page pada link-list setiap memory reference
  - Counter untuk tiap page yang di-update secara otomatis untuk setiap memory reference
- Contoh:

|                                                                        |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 2                                                                      | 3 | 2 | 1 | 5                                                                       | 2 | 4 | 5 | 3                                                                       | 2 | 5 | 2 |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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|                                                                        |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
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| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 1                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 1                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 1                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 4                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 4                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 4                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 3                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 5                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
| 2                                                                      |   |   |   |                                                                         |   |   |   |                                                                         |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |
|                                                                        |   |   |   | F                                                                       |   | F |   | F                                                                       | F |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |                                                                          |   |   |   |

- Jumlah page-fault untuk penggantian page = 4

# **Storage Media (Media Penyimpanan)**

# Gambaran dari Magnetic Disk

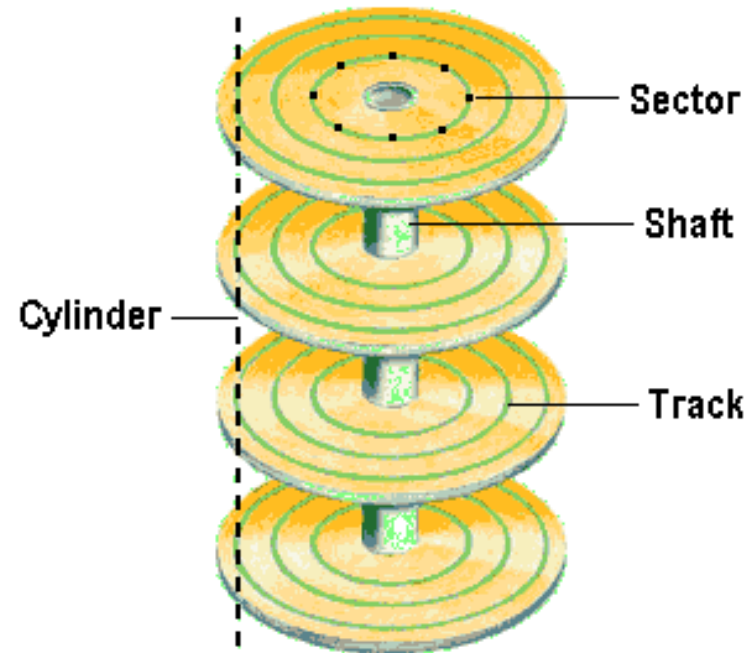


## Standard Koneksi external

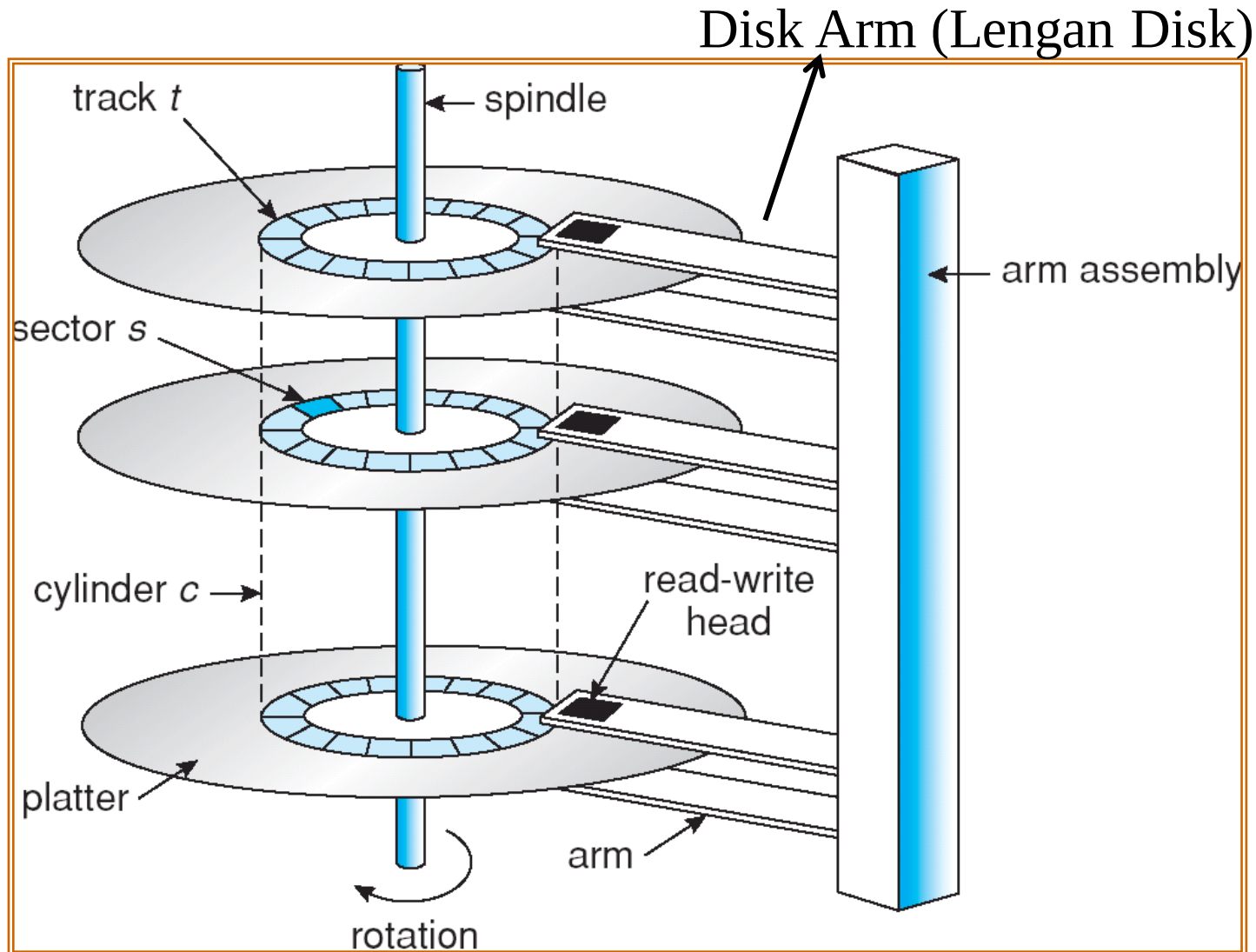
- IDE/ATA/SATA
- SCSI
- USB

# Harddisk Geometri

- **Platters**
  - Lempengan material magnetic 2 sisi
    - 2 heads untuk read/write
  - Kecepatan putar 3000-15000 RPM
- **Tracks**
  - Lingkaran concentric
  - Dibagi atas beberapa sectors
- **Cylinders**
  - Track yang sama pada tiap plater
- **Operasi**
  - Seek: gerakkan arm ke track
  - Read/write: tunggu sampai sector berada di bawah read/write head, kemudian transfer data
- Controller konversikan physical address [cylinder, head, sector] menjadi logical disk address berupa array of sectors atau blocks(group of sector)
- [cylinder.head.sector] → CHS addressing



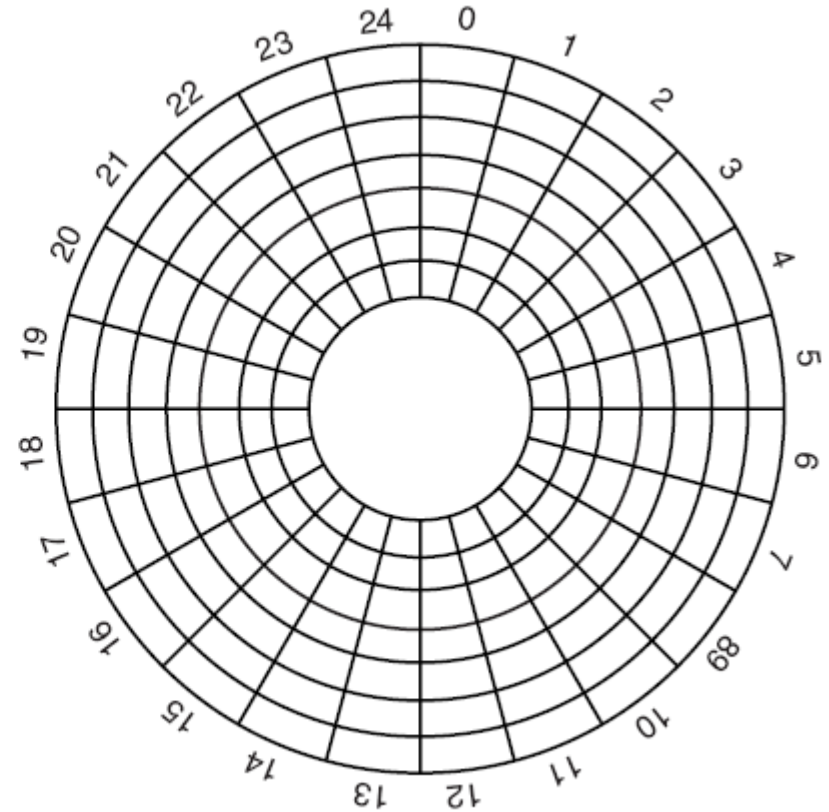
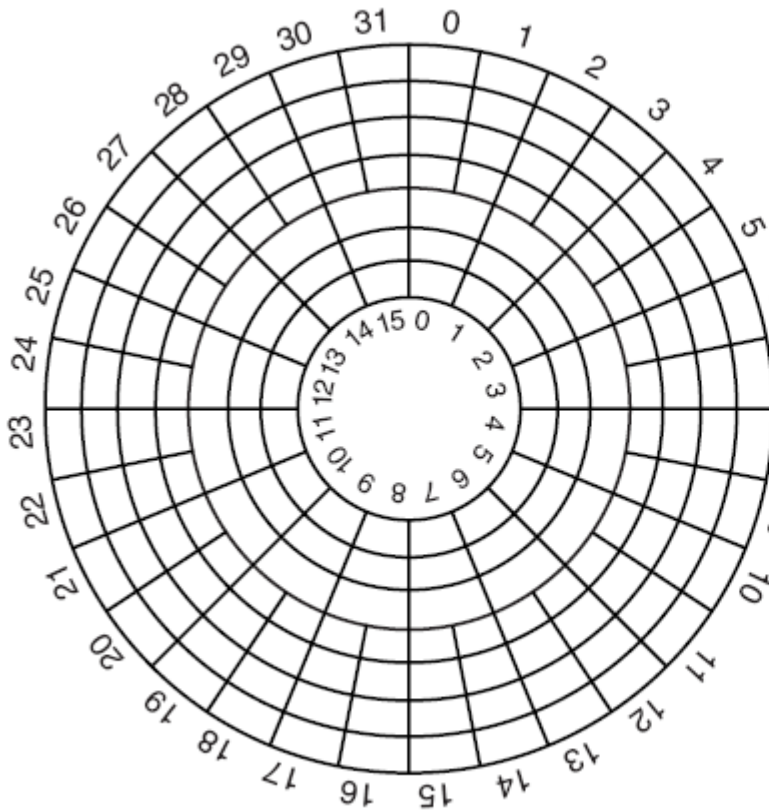
# Pergerakan Head (Read/Write)



Ref: WPI Lecture notes

**[cylinder,head,sector] → CHS addressing**

# Harddisk Geometri



**(a) Physical geometry of a disk with two zones.**

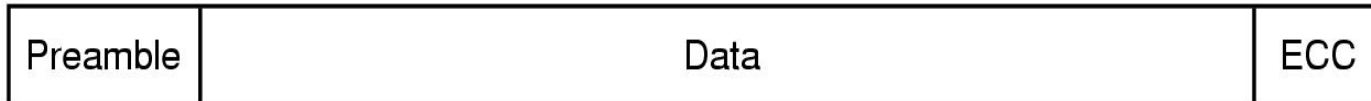
**(b) A possible virtual geometry for this disk.**

→ logical sector/block

# Disk Format (1)

---

- Sebelum digunakan, disk harus di-format
- Low-level format
  - Susun track pada tiap platter dan bagi atas beberapa sectors. Letakkan celah sempit antar sector, dan juga siapkan spare sectors (sector cadangan). Gambaran suatu sector adalah sbb.:



- Preamble: bit pattern untuk deteksi awal dari suatu sector. Juga berisi informasi cylinder dan nomor sector, dan informasi untuk sinkronisasi
- Data: ukuran ditentukan saat di-format
- ECC: error correction code untuk memastikan tidak ada kesalahan. Algoritma yang digunakan tergantung pembuat hardisk

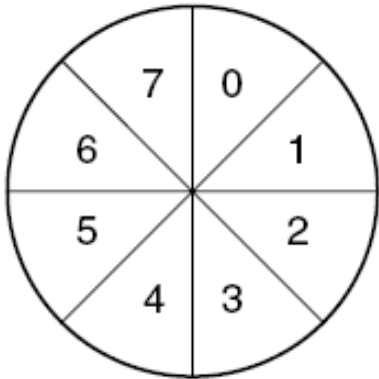
# Disk Format (2)

---

- **Lakukan partisi pada harddisk**
  - **Buat partition table**
  - **Seolah-olah ada beberapa disk yang tersedia yang dapat digunakan oleh File System yang berbeda**
- **High-level format pada tiap partisi**
  - **Buat boot block**
  - **Susun administrasi free blocks**
  - **Tentukan file system dan root directory**

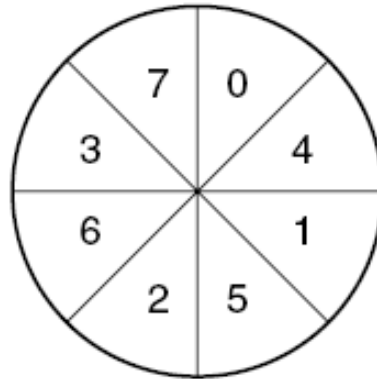
# Disk Format: Interleaving

- Penomoran sektor yang melompat (tidak berurutan), untuk memberi waktu atau kesempatan pada controller untuk transfer data ke memory



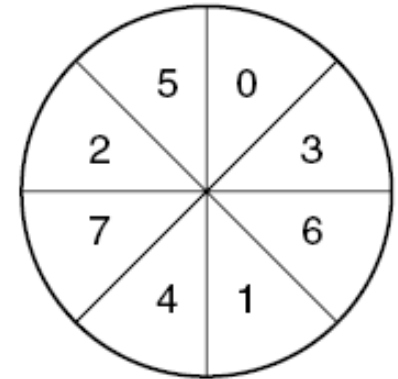
(a)

(a) No interleaving.



(b)

(b) Single interleaving.



(c)

(c) Double interleaving.

# **Disk Arm Scheduling**

## **(Penjadwalan Lengan Disk)**

---

- Waktu yang diperlukan untuk read/write disk block:
  - **Seek time:** waktu untuk menggerakkan disk arm ke target cylinder
  - **Rotational Delay:** waktu berputarnya platter sampai target sector berada di bawah read/write head
  - **Data transfer time**
- **Delay didominasi oleh seek time**
- **Disk scheduling algorithm adalah algoritma untuk meminimalkan disk seek time dan mengusahakan fairness (keadilan) atas prioritas permintaan read/write disk block**
- **Disk drives mengelola tabel permintaan read/write dengan nomor cylinder sebagai index**

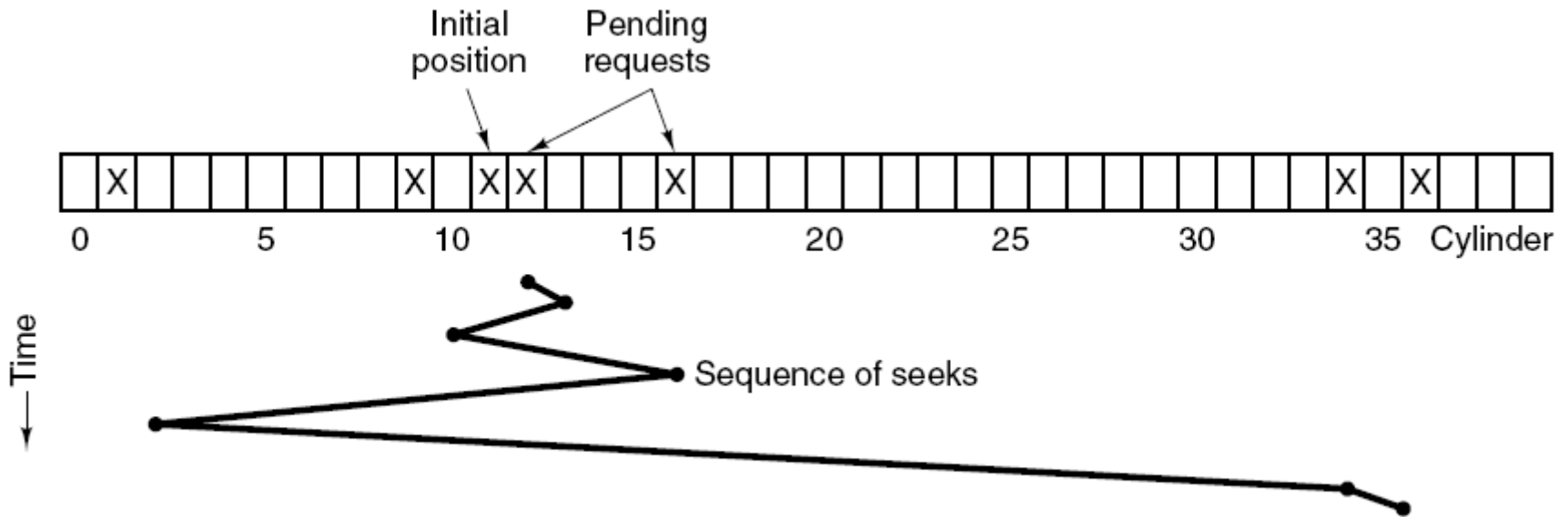
# First Come First Served (FCFS) Disk Scheduling Algorithm

---

**Contoh:**

- **Posisi awal = 11. Urutan permintaan: 1, 36, 16, 34, 9, 12**
- **Pergerakan: 11 → 1 → 36 → 16 → 34 → 9 → 12**
- **Total pergerakan arm =  $10+35+20+18+25+3 = 111$  cylinders**
- **Perlu upaya untuk mengurangi total pergerakan arm, untuk mengurangi seek time**

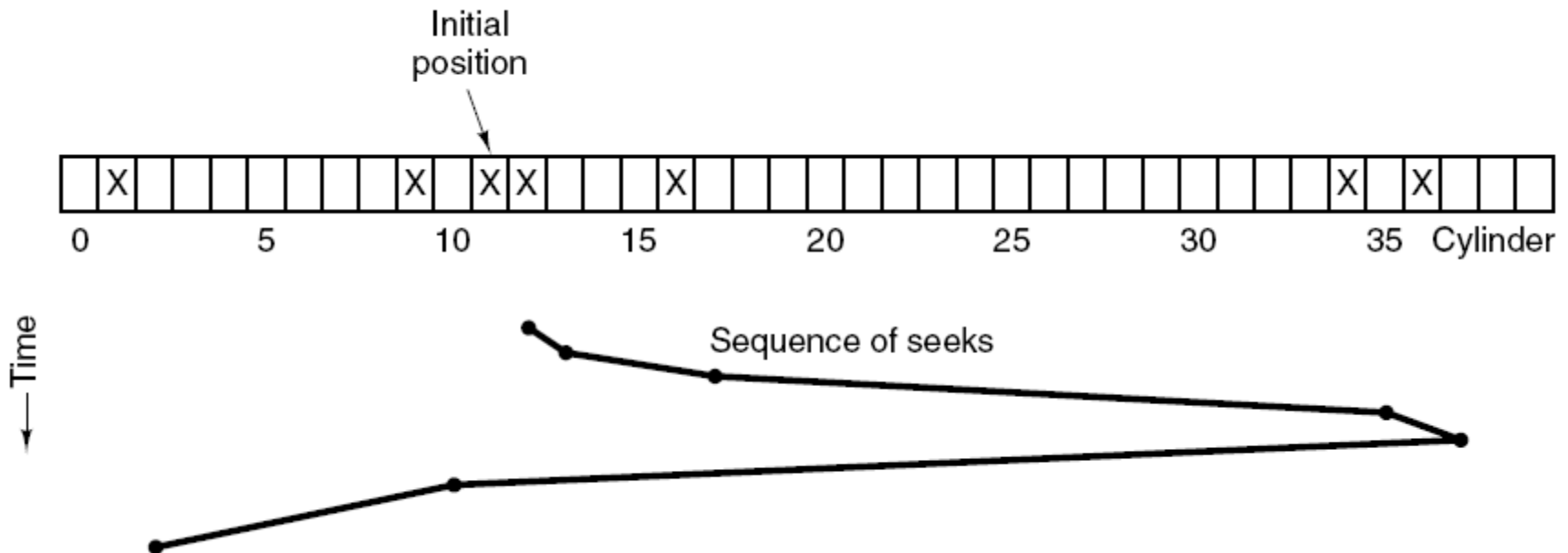
# Shortest Seek First (SSF) Disk Scheduling Algorithm



- Kalau gunakan Shortest Seek First (SSF), total pergerakan arm =  $1+3+7+15+33+2 = 61$  cylinders → jauh lebih cepat dibandingkan FCFS
- Kekurangan: saat disk loading tinggi, ada kecenderungan disk arm untuk selalu berada di tengah sehingga tidak adil untuk permintaan block yang ada di pinggir

# Elevator atau SCAN Algorithm

- Contoh: posisi awal = 11. Urutan permintaan: 1, 36, 16, 34, 9, 12
- Elevator algorithm:
  - Mengatasi konflik target minimum seek time dan fairness
  - Disk arm melanjutkan pergerakan sesuai arah (up atau down) sampai tidak ada lagi permintaan di arah tersebut, baru berbalik arah



- Total pergerakan arm =  $1+4+18+2+27+8 = 60$  cylinders

# Seek Time

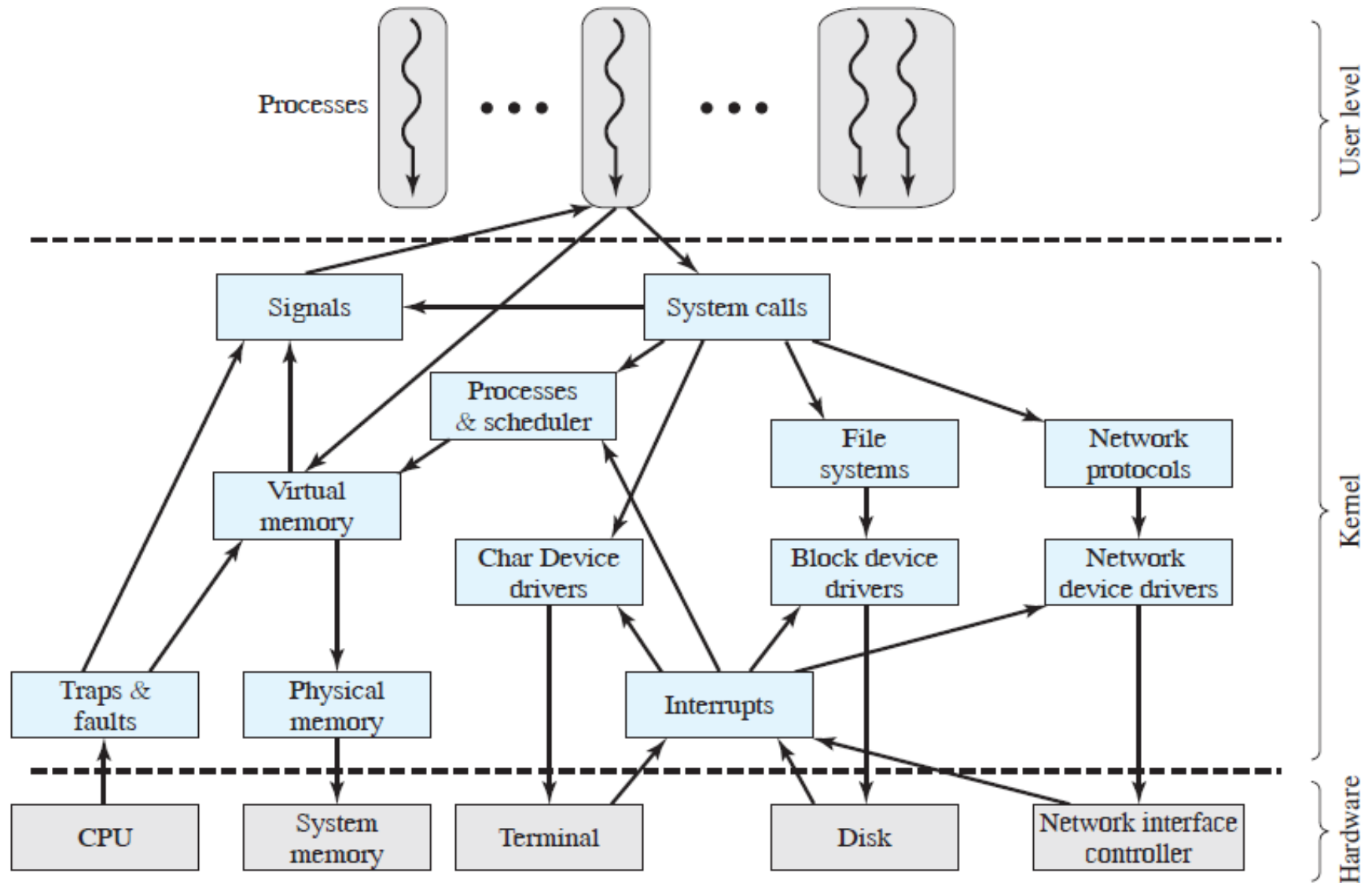
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- Jika diketahui seek time = 5 msec/cylinder (waktu yang diperlukan untuk memindahkan lengan disk tiap satu silinder), maka seek time keseluruhan dari masing-masing algoritma di slide sebelumnya adalah:
- FCFS:  $111 * 5 = 555$  msec
- SSF:  $61 * 5 = 305$  msec
- Elevator:  $60 * 5 = 300$  msec

# **File Systems**

## **(Sistem-sistem Berkas)**

# Linux Kernel Components



# **File Systems**

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- **Suatu sistem komputer memerlukan media penyimpanan informasi di mana**
  - Memungkinkan menyimpan informasi yang besar
  - Bertahan lama dan tetap ada walau process yang membuat/menggunakannya telah berakhir
  - Beberapa process dapat serentak menggunakannya
- **OS berperan dalam hal**
  - Abstraksi berupa FILE sehingga ada suatu interface yang sama dalam pengaksesan informasi pada suatu media, misal penamaan file, attribute, dan system calls yang sama
  - Pengelolaan resources → format file, pengaksesan informasi, alokasi disk blocks, media format, proteksi, dsb.
- **File System adalah bagian dari Operating System yang bertugas dalam pengelolaan file**
  - Pandangan pengguna (interface)
  - Implementasi

# **File Naming (Penamaan File)**

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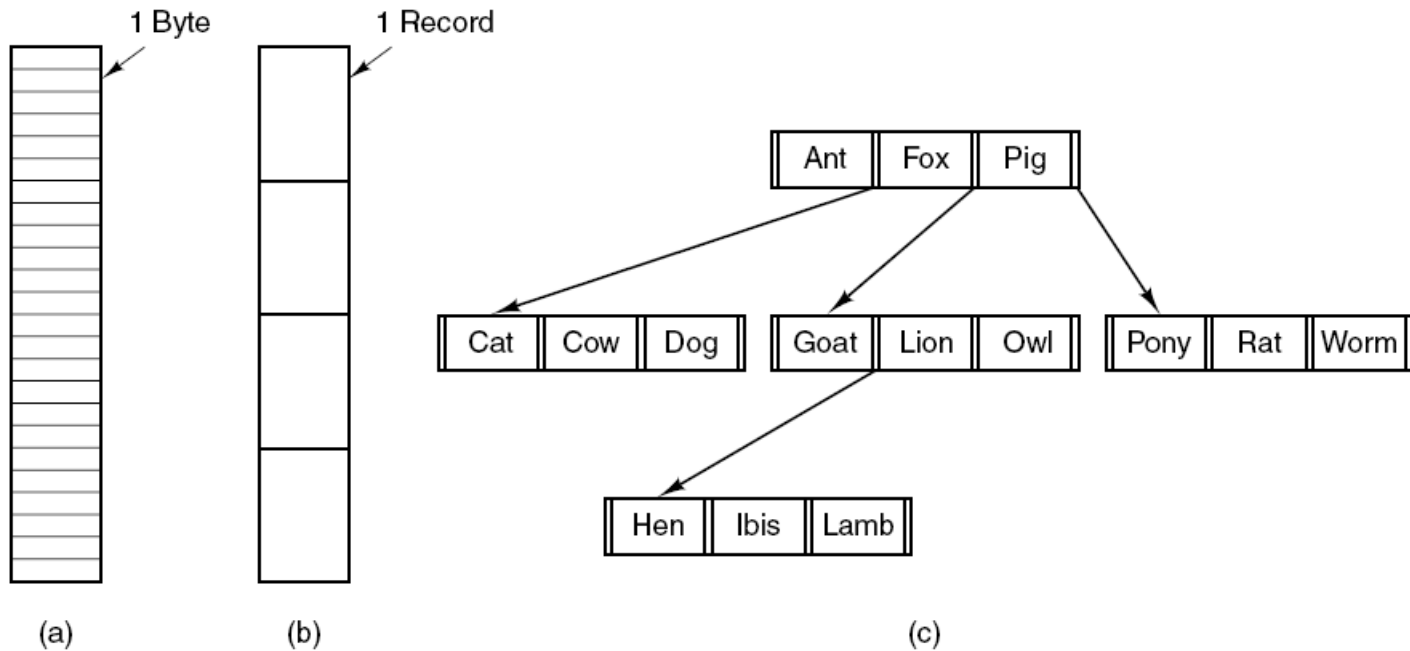
- **File dibuat dan diberikan nama oleh suatu process. Saat process berakhir, file tetap ada di media penyimpanan dan process yang lain dapat mengakses kembali dengan menggunakan namanya**
- **Aturan penamaan bervariasi tergantung file system yang digunakan**
  - **Umumnya sudah mendukung penggunaan 255 karakter**
  - **Case sensitif atau tidak**
  - **Gunakan extension atau tidak, dsb**

# File Naming

---

| Extension | Meaning                                           |
|-----------|---------------------------------------------------|
| file.bak  | Backup file                                       |
| file.c    | C source program                                  |
| file.gif  | Compuserve Graphical Interchange Format image     |
| file.hlp  | Help file                                         |
| file.html | World Wide Web HyperText Markup Language document |
| file.jpg  | Still picture encoded with the JPEG standard      |
| file.mp3  | Music encoded in MPEG layer 3 audio format        |
| file.mpg  | Movie encoded with the MPEG standard              |
| file.o    | Object file (compiler output, not yet linked)     |
| file.pdf  | Portable Document Format file                     |
| file.ps   | PostScript file                                   |
| file.tex  | Input for the TEX formatting program              |
| file.txt  | General text file                                 |
| file.zip  | Compressed archive                                |

# File Structure (Struktur File)



(a) Byte sequence. (b) Record sequence. (c) Tree.



Digunakan oleh MSDos, Windows versi awal, Unix, dsb.  
OS tidak berusaha menginterpretasikan suatu file, tapi dilakukan oleh user-level program

# **File Types dan File Access**

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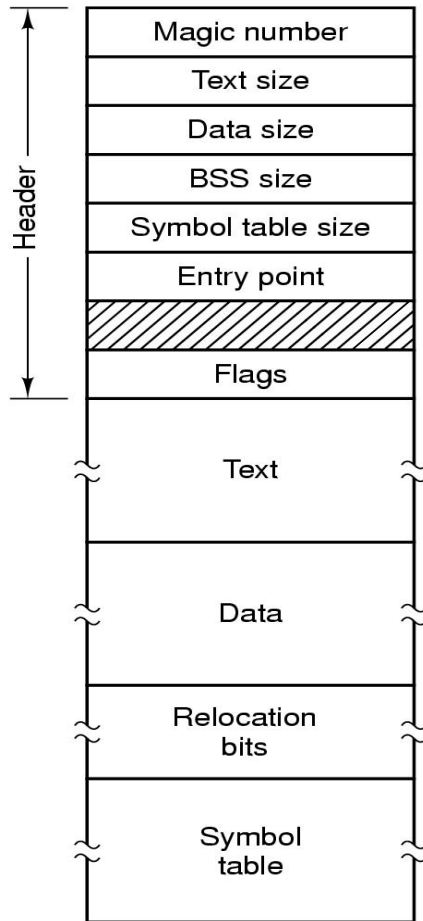
- **File Types**

- **Regular file**
  - ASCII File
  - Binary File
- **Directory/Folder: system file untuk menata struktur dari File System**
- **Special Files, contoh pada Unix OS:**
  - Character Special File → untuk pemodelan perangkat I/O serial
  - Block Special File → untuk pemodelan disk

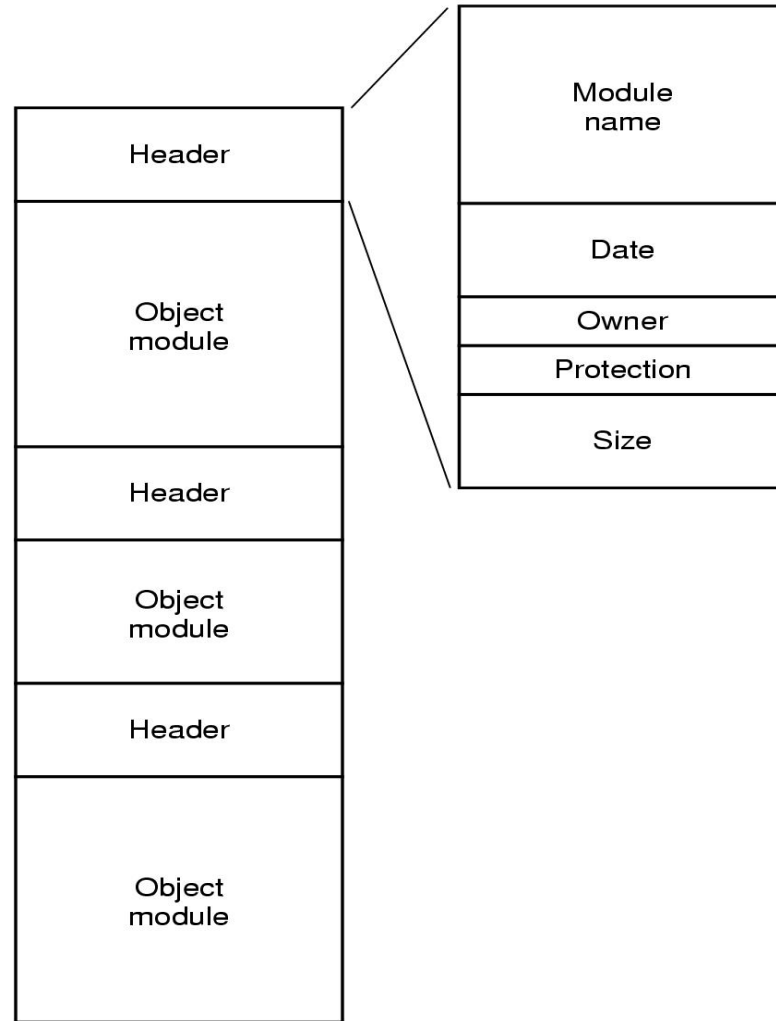
- **File Access**

- Sequential Access
- Random Access

# Contoh binary files



(a)



(b)

(a) An executable file. (b) An archive.

# File Attributes / Metada

- Informasi tambahan dari suatu File

| Attribute           | Meaning                                               |
|---------------------|-------------------------------------------------------|
| Protection          | Who can access the file and in what way               |
| Password            | Password needed to access the file                    |
| Creator             | ID of the person who created the file                 |
| Owner               | Current owner                                         |
| Read-only flag      | 0 for read/write; 1 for read only                     |
| Hidden flag         | 0 for normal; 1 for do not display in listings        |
| System flag         | 0 for normal files; 1 for system file                 |
| Archive flag        | 0 for has been backed up; 1 for needs to be backed up |
| ASCII/binary flag   | 0 for ASCII file; 1 for binary file                   |
| Random access flag  | 0 for sequential access only; 1 for random access     |
| Temporary flag      | 0 for normal; 1 for delete file on process exit       |
| Lock flags          | 0 for unlocked; nonzero for locked                    |
| Record length       | Number of bytes in a record                           |
| Key position        | Offset of the key within each record                  |
| Key length          | Number of bytes in the key field                      |
| Creation time       | Date and time the file was created                    |
| Time of last access | Date and time the file was last accessed              |
| Time of last change | Date and time the file was last changed               |
| Current size        | Number of bytes in the file                           |
| Maximum size        | Number of bytes the file may grow to                  |

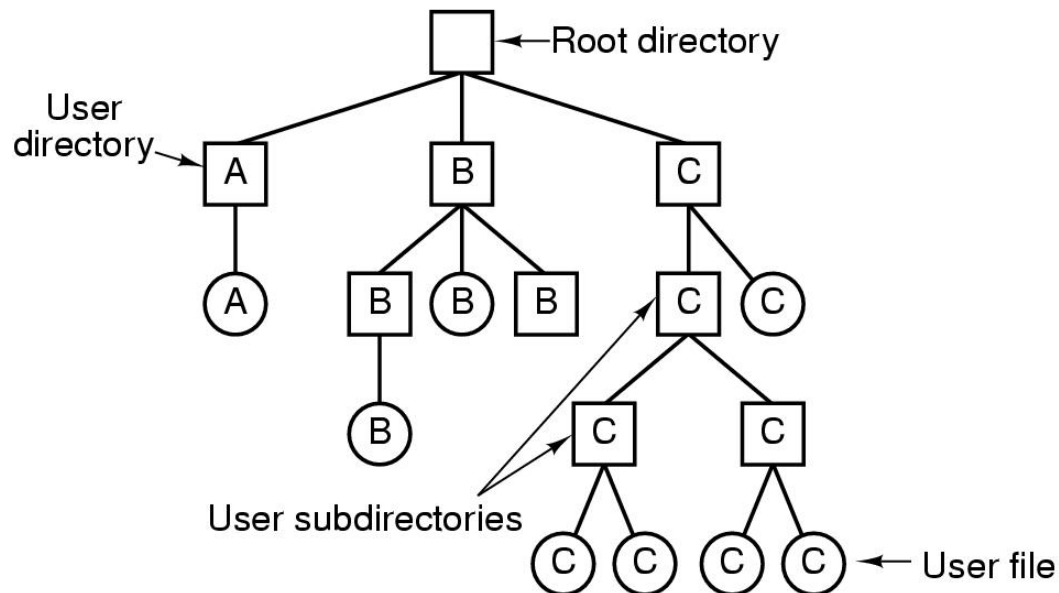
# **File Operations**

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- **Create:** membuat file dengan attributes, tanpa data
- **Delete:** menghapus file dari disk
- **Open:** membuka file sebelum digunakan oleh process. File attributes dan disk address dibawa ke memory
- **Close:** menutup file, dan file table space dikosongkan
- **Read:** membaca isi file
- **Write:** menulis data ke file
- **Append:** bentuk khusus dari write, dimana data ditulis di akhir dari file
- **Seek:** memposisikan file pointer ke posisi yang spesifik di file
- **Get Attributes:** mendapatkan file attributes
- **Set Attributes:** menetapkan file attributes
- **Rename:** mengubah nama file

# Directory/Folder

- **Single-level directory Systems**
  - Hanya ada satu directory yaitu root
  - Sederhana dan cepat dalam mencari lokasi file
  - Penggunaan pada embedded system untuk aplikasi spesifik/terbatas, mis.: digital camera, mp3 player, dsb
- **Hierarchical Directory Systems**
  - Untuk general purpose computer, dengan jumlah file yang besar, multi-user, dsb.
  - Contoh pengelompokan file berdasarkan directory:



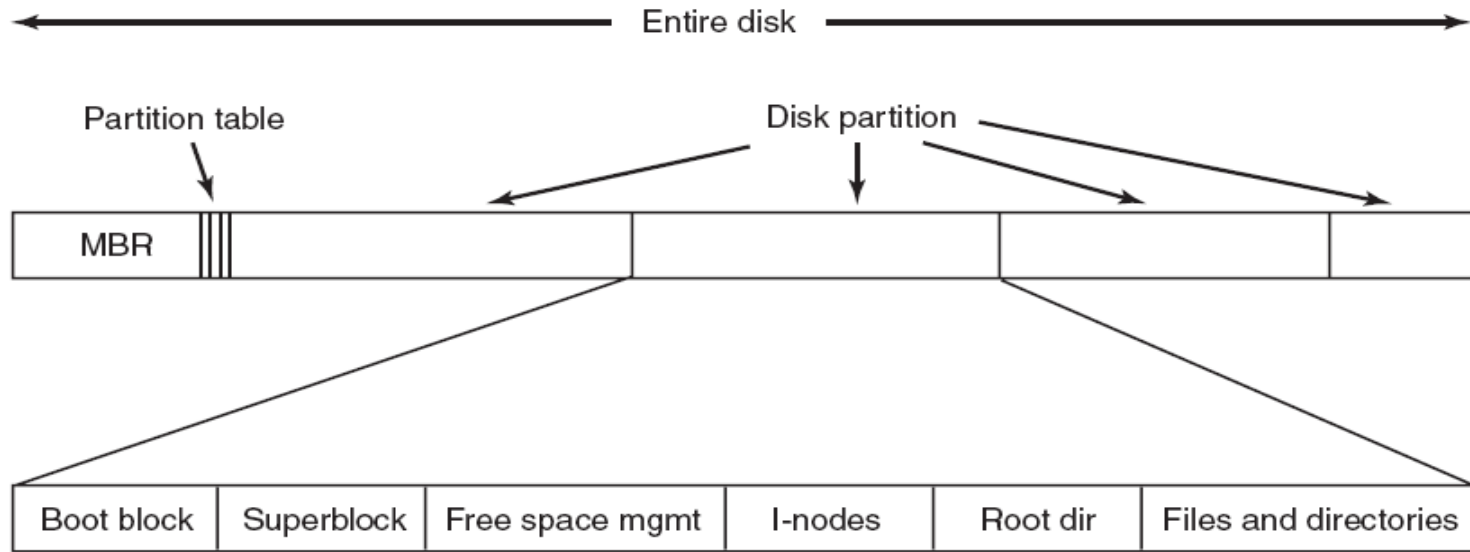
# **File System Implementation**

## **(Implementasi Sistem Berkas)**

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- **File System Layout**
- **File Allocation**
- **Journalling File System**
- **Virtual File System**

# File System Layout



MBR: Master Boot Record, lokasi pada disk sektor 0

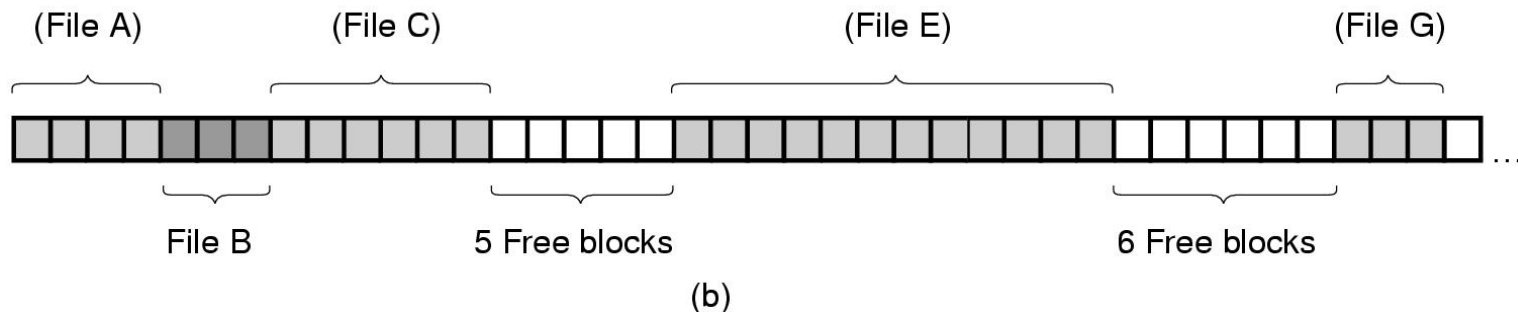
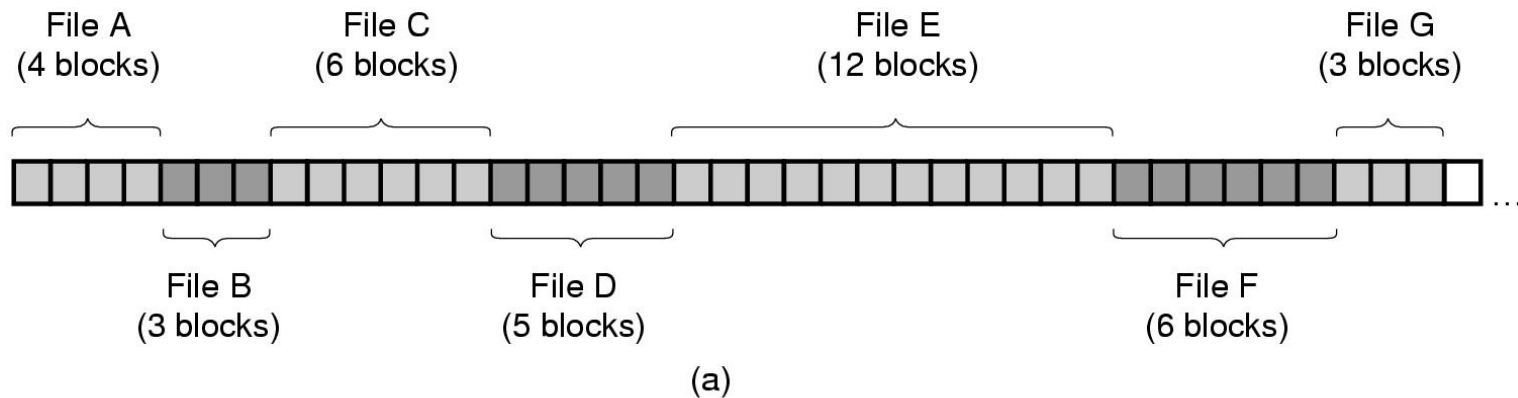
Partition table berisikan address awal dan akhir dari tiap partisi

Layout partisi bervariasi tergantung file system yang digunakan

- Boot block load OS yang ada di partisi
- Superblock berisikan parameter dari file system (mis. Jenis file system, jumlah blok, dsb.) yang di-load ke memory

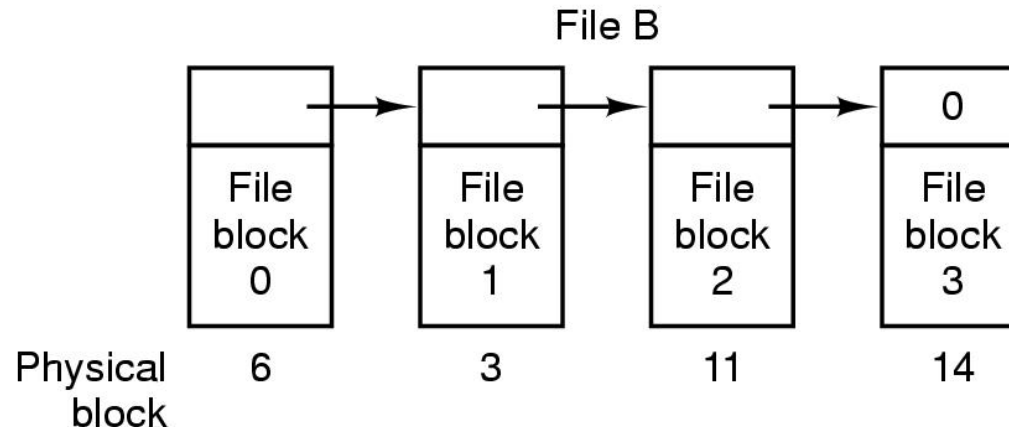
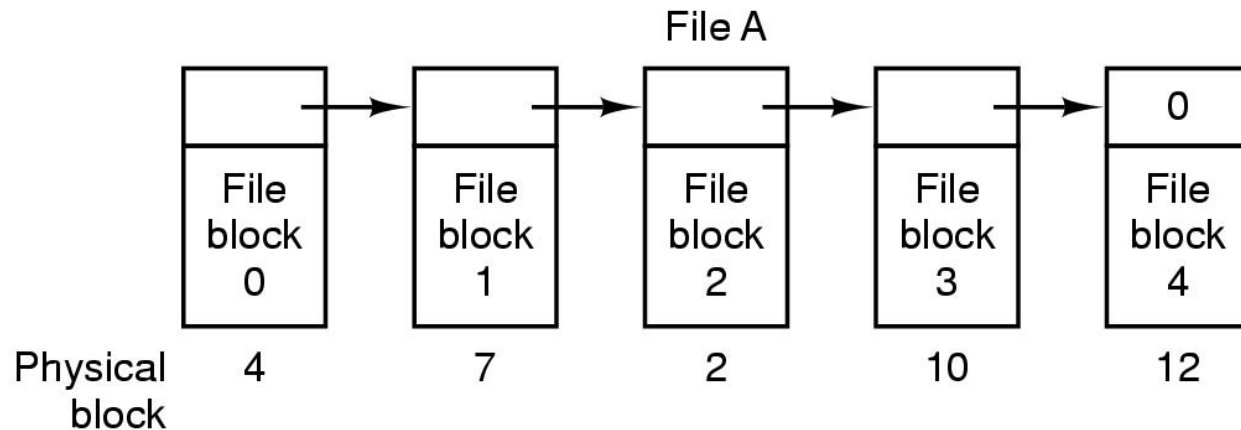
# File Allocation: Contiguous Allocation

- Alokasi file pada blok disk yang berurutan
- Keuntungan: mudah diimplementasikan
- Kekurangan: potensi fragmentation
- Idealnya, perlu tahu ukuran file lebih dahulu sebelum dialokasikan.  
Cocok untuk write-once optical media, mis. CDRom, DVD.



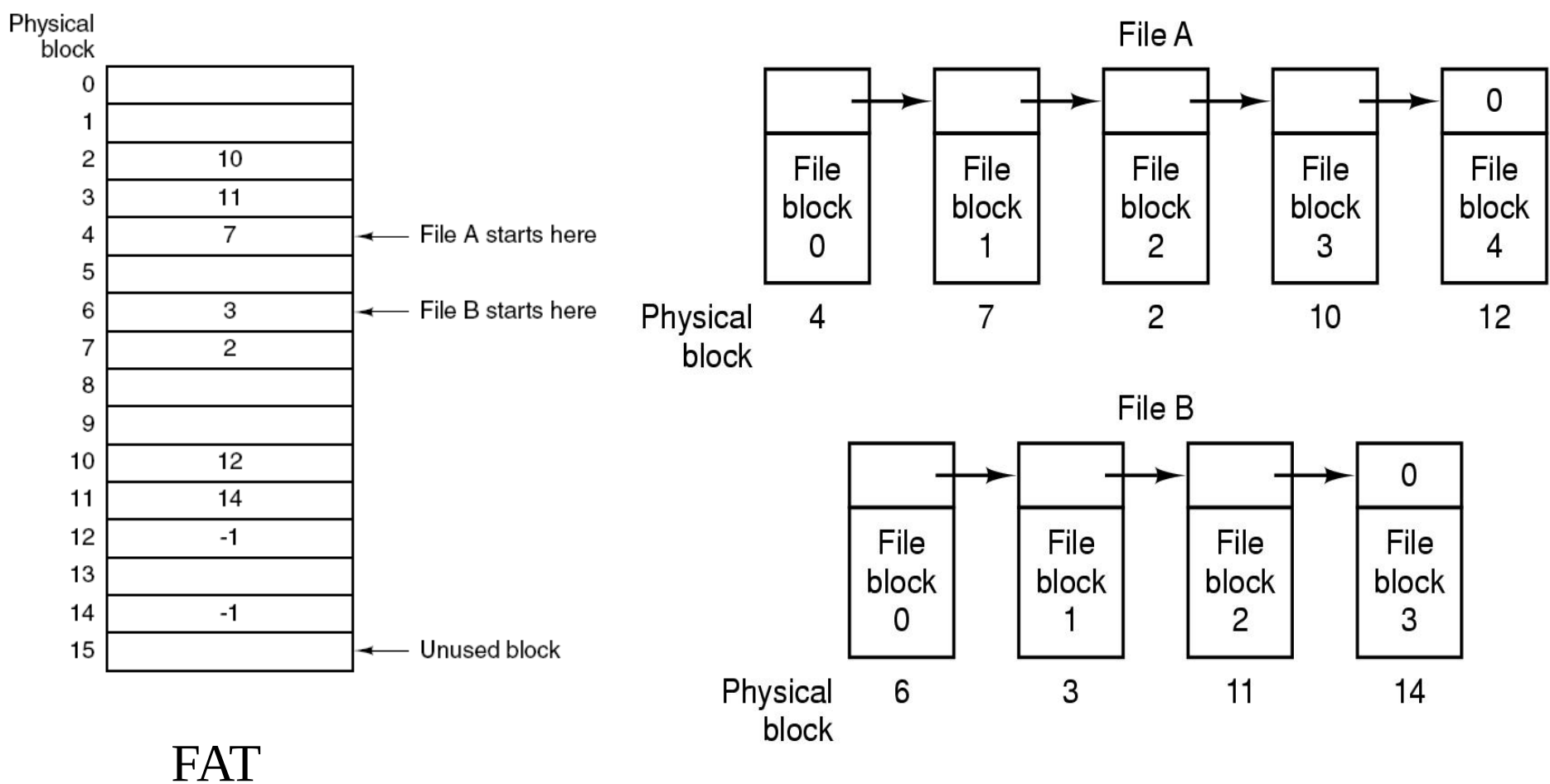
# File Allocation: Link-List Allocation

- Directory entry hanya perlu menyimpan disk address dari blok pertama
- Mengurangi potensi fragmentasi karena alokasi tidak harus pada blok yang bersebelahan, tapi random access menjadi lambat



# File Allocation: Link-List Allocation menggunakan Tabel di Memory

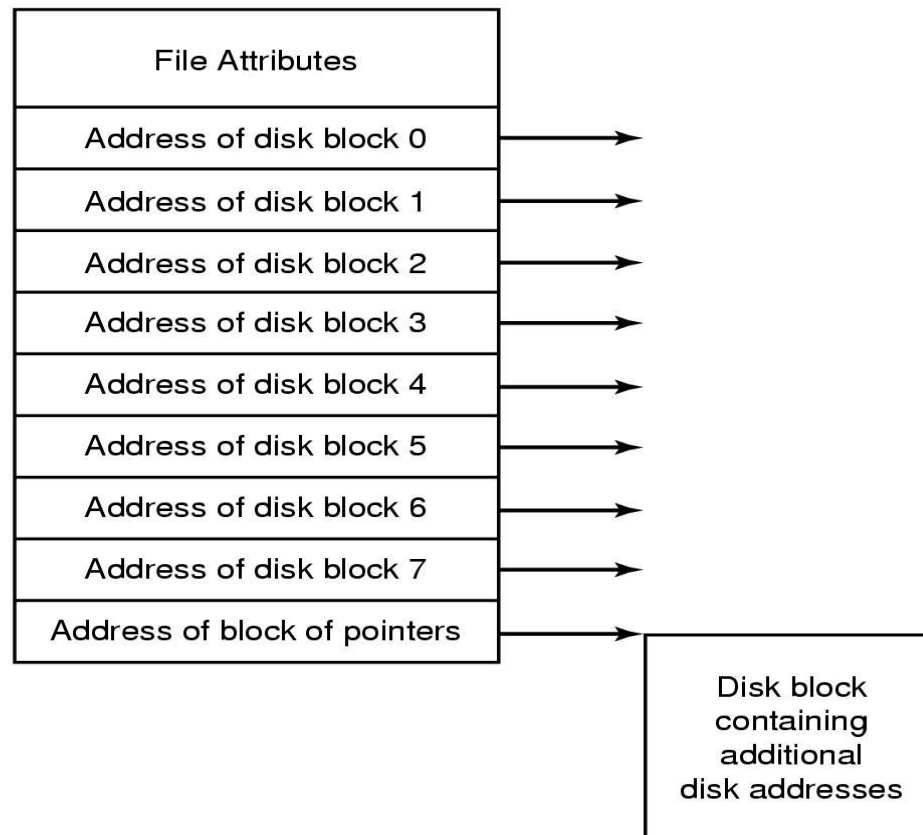
- **Pointer untuk tiap disk block disimpan pada FAT (File Allocation Table) yang ada di memory, sehingga random access menjadi lebih cepat**



# File Allocation: i-Node (index Node)

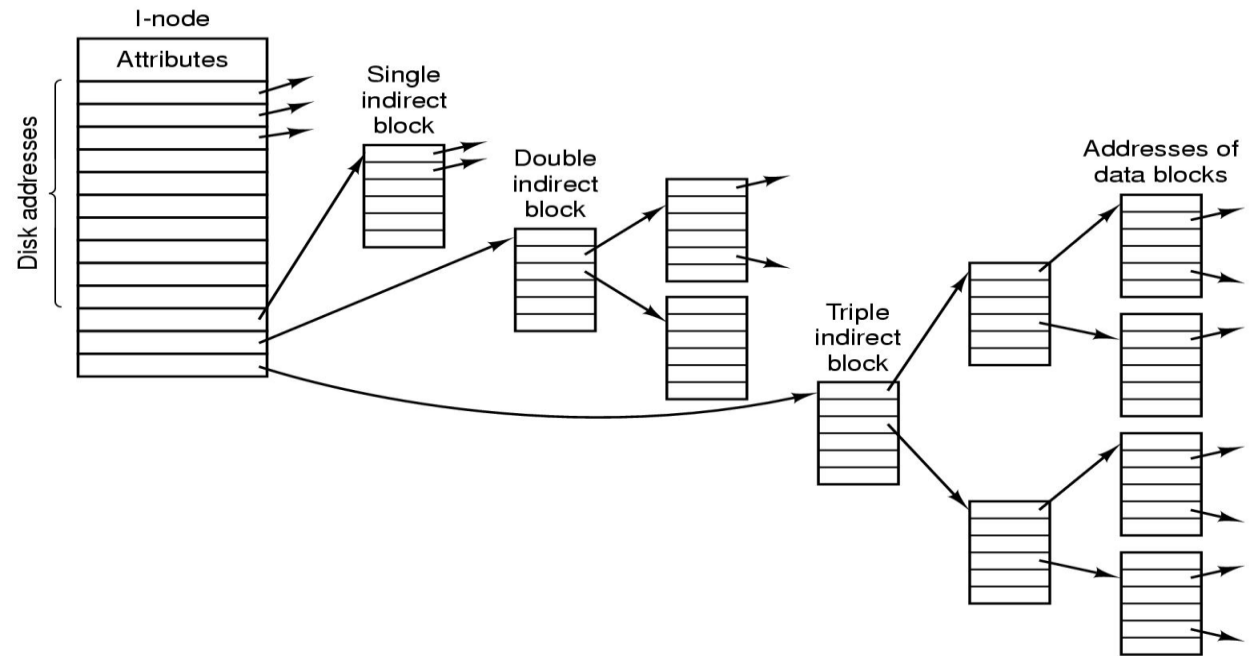
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- Asosiasikan tiap file dengan sebuah struktur data berisikan file attributes dan address dari disk block
- Keuntungan: i-Node hanya perlu ada di memory hanya apabila file yang diasosiasikan dibuka (open)



# Contoh: Unix/Linux File Systems

- Contoh: 4-byte disk address, block size = 512 bytes, 10 direct address, 1 single indirect block, 1 double indirect block, dan 1 tripple indirect block
- Single indirect  $\rightarrow 512/4 = 128$  addresses
- Double indirect  $\rightarrow 128 \times 128 = 16384$  addresses (16K blocks)
- Triple indirect  $\rightarrow 128 \times 128 \times 128 = 2097152$  addresses (2M blocks)
- Max file size  $\approx 10 \times 0.5\text{KB} + 128 \times 0.5\text{K} + 16\text{K} \times 0.5\text{K} + 2\text{M} \times 0.5\text{K} = 5\text{KB} + 64\text{KB} + 8\text{MB} + 1\text{GB}$



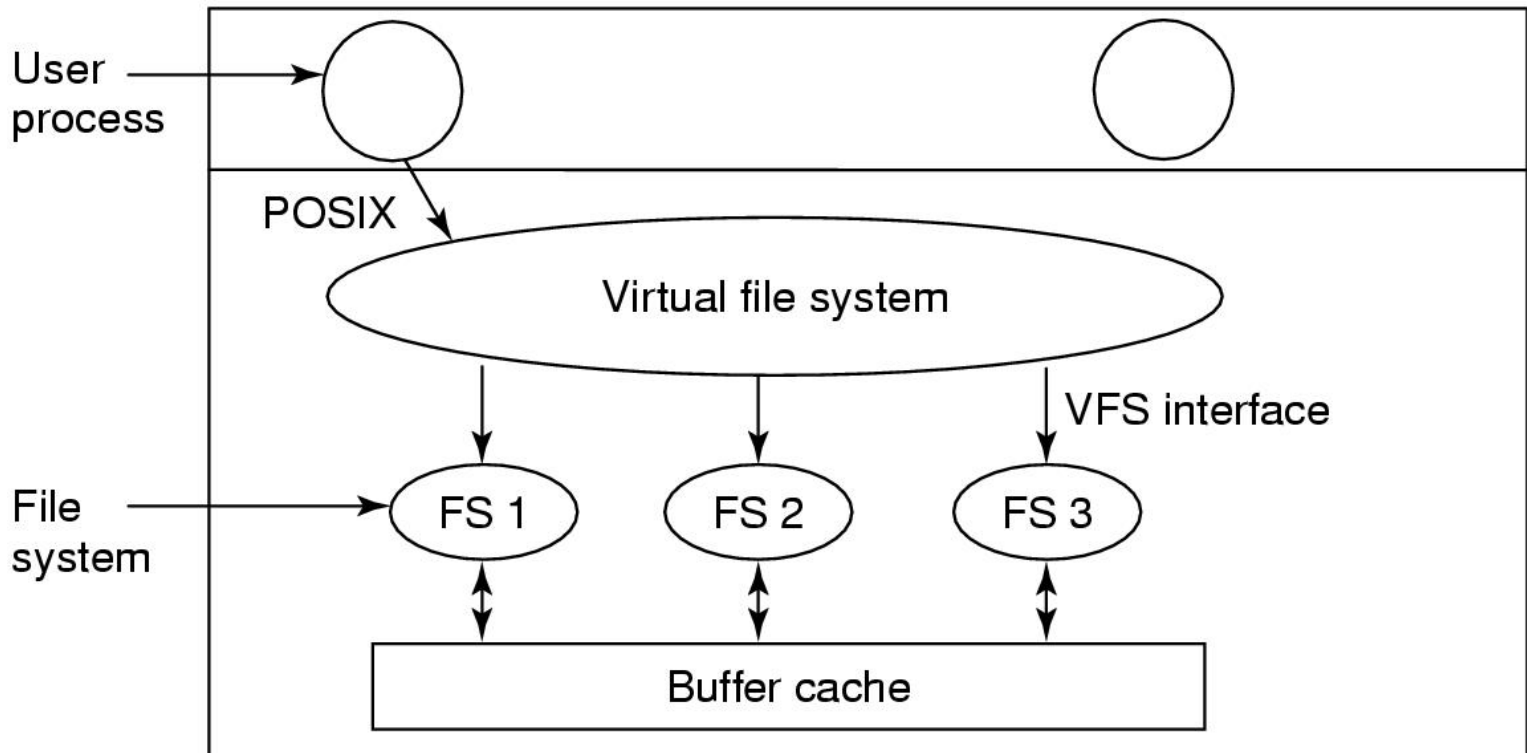
# **Journalling File Systems**

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- **Mekanisme di mana sistem menyimpan informasi ke dalam log file (journal) tentang apa yang sistem akan lakukan. Sehingga apabila sistem crash di mana tugas belum selesai dijalankan, maka setelah reboot dapat dilanjutkan**
- **Contoh dalam menghapus file:**
  - **Hapus file dari directory**
  - **Kembalikan i-Node ke himpunan free i-Nodes (i-Nodes yang tidak terpakai)**
  - **Kembalikan semua disk blocks ke himpunan free disk blocks (disk blocks yang tidak terpakai)**
  - **Bagaimana kalau crash terjadi saat salah satu dari ketiga langkah di atas belum selesai dijalankan?**

# Virtual File System (VFS)

- Mekanisme penyatuan beberapa file system menjadi satu struktur sehingga tampak transparan dari sisi pengguna



# Akhir Kuliah Minggu 11

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**Terima kasih atas perhatiannya!**